

SFC-215 — Independent Structural Convergence

Why Structural Duality May Strengthen the Engineering Interpretation of LLMs

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Repository: <https://github.com/sizhet/Structural-Feasibility-Confidence-SFC>

Abstract

The interpretation of Large Language Models (LLMs) has become one of the central questions in modern AI research.

This repository proposes an engineering interpretation in which many LLM capabilities can be understood as forms of **implicit structural feasibility evaluation**.

This article does not attempt to prove that interpretation. Instead, it asks a different methodological question:

Why should such an interpretation be taken seriously?

We argue that its credibility is strengthened by a phenomenon we call **Independent Structural Convergence**.

When two theories developed largely independently repeatedly converge on the same engineering phenomena, the convergence itself becomes an additional source of engineering confidence.

This confidence does not constitute proof.

Rather, it provides an independent engineering prior that motivates further investigation, empirical testing, and theoretical refinement.

1. Engineering Confidence Does Not Arise from a Single Source

Engineering rarely waits for complete proof.

Bridges, aircraft, operating systems, and industrial control systems are often developed by combining multiple partially independent sources of evidence.

The confidence of an engineer grows not because one argument proves everything, but because different arguments point toward the same conclusion.

Scientific theories frequently evolve in a similar manner.

Independent observations, mathematical analyses, engineering practices, and experimental evidence gradually reinforce one another. The resulting confidence emerges from convergence rather than from any single line of reasoning.

2. Structural Intelligence Was Not Developed to Explain LLMs

One important characteristic of the Structural Intelligence (SI) research program is historical.

The core concepts—including Calling Graphs, Differential Trees, Function Tunnels, Structural Cognitive Runtime, Gap Bridging, and Structural Feasibility—were introduced to address engineering and structural computation problems rather than to interpret Transformer architectures. Only after these structural frameworks had matured did an unexpected observation appear.

Many practical behaviors exhibited by modern LLMs seemed increasingly compatible with concepts that had originally been developed for entirely different purposes.

This historical independence is significant.

It reduces the possibility that the structural interpretation merely restates concepts already embedded within existing LLM theories.

3. Structural Duality

The relationship between SI and LLMs may therefore be viewed as a form of **Structural Duality**.

The two frameworks originate from different assumptions and development paths.

One begins with explicit structural engineering principles.

The other emerges through large-scale statistical optimization over human-generated data.

Despite these differences, both increasingly appear to address similar classes of engineering problems.

Examples include:

- partial solution construction,
- engineering gap bridging,
- feasibility evaluation,
- iterative refinement,
- structural completion,
- multi-step engineering reasoning.

Structural Duality does not imply that the two systems implement identical

algorithms.

Rather, it suggests that different computational mechanisms may converge toward similar structural behaviors.

4. Independent Structural Convergence

When independently developed theories repeatedly explain the same engineering phenomena, a new kind of evidence begins to emerge.

This evidence is neither deductive nor statistical.

Instead, it arises from structural convergence.

In the present context:

- the SI framework predicts that explicit structural mechanisms can support complex engineering reasoning;
- modern LLMs increasingly demonstrate engineering behaviors that are compatible with those mechanisms;
- and the Structural Feasibility framework offers a common language capable of describing both.

This convergence does not prove that LLMs internally implement the same structures proposed by SI.

However, it increases the plausibility that both frameworks are capturing important aspects of the same broader class of structural computation.

5. Structural Benefit of Doubt

Engineering often proceeds by assigning reasonable confidence to hypotheses that are supported by multiple independent sources.

We propose that a similar principle applies here.

If an independently developed structural theory successfully explains increasingly broad engineering phenomena, then observations consistent with that theory naturally receive additional engineering credibility.

We refer to this additional credibility as a **Structural Benefit of Doubt**.

Importantly, this benefit does **not** arise from preference for one theory over another.

Nor does it imply that competing interpretations are incorrect.

Instead, it reflects the practical engineering observation that independent convergence generally increases confidence while leaving room for future revision.

6. An Invitation Rather Than a Conclusion

This article should not be interpreted as a claim that Structural Intelligence has already explained the nature of LLMs.

Such a claim would require substantially more theoretical analysis and

empirical evidence.

Instead, this work proposes a methodological perspective.

Independent theories that repeatedly converge on the same engineering behaviors deserve careful attention because they provide complementary explanatory viewpoints.

Whether Structural Duality ultimately proves to be fundamental remains an open research question.

The purpose of this article is therefore not to close discussion, but to encourage it.

We welcome alternative explanations, critical analysis, empirical challenges, and theoretical improvements.

If future evidence supports these ideas, Independent Structural Convergence may become an important methodological principle for understanding AI.

If future evidence contradicts them, the discussion itself will still have contributed to clarifying the structural questions that next-generation AI research must answer.

Concluding Remarks

Scientific progress is often driven not by the victory of one theory over another, but by the gradual convergence of multiple independent lines of inquiry.

Perhaps the greatest value of Structural Duality is not that it immediately proves an engineering interpretation of LLMs.

Its value lies in demonstrating that independently developed structural theories may illuminate the same engineering phenomena from different directions.

Such convergence provides neither certainty nor finality.

It provides something equally important for scientific progress:

a principled reason to continue asking better questions.